Opening your eyes wide allows you to see more clearly

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In this study, we address a fundamental hardware-level bottleneck in human visual perception: eyelid occultation. We observe that the acquisition of visual data is strictly proportional to the physical diameter of the palpebral fissure. To address this, we introduce the Active Eyelid Elevation algorithm, a systematic framework for biological shutter modulation that ensures a continuous high-fidelity video stream. By implementing the AEE protocol, we successfully resolve the long-standing issue of human agents being unable to identify roommates stealing snacks during sleep-states. Our experimental results demonstrate that under optimal lighting, the AEE-driven "Open" state outperforms the "Closed" baseline by approximately 100%. This finding provides a robust physical foundation for solving the "Blindness-by-Choice" challenge in everyday life.

1 Introduction

In the field of computer vision, researchers often focus on the depth of neural networks while ignoring the physical constraints of the primary sensor. If the eyelids remain in a state of "Downwards Occlusion," [3, 5, 8, 7, 1] even the most advanced post-processing algorithms fail to extract features from a zero-photon input stream. This hardware-level signal interruption is commonly referred to in the literature as "being asleep". [1]

The core hypothesis of this paper is straightforward: by manually or biologically contracting the *levator palpebrae superioris* muscle, one can forcibly expand the field of view. This is not merely a mechanical process, but a critical semantic alignment procedure. [2, 6, 4]

Despite the intuitive benefits of observation, several critical **challenges** hinder robust visual acquisition: 1) **Stochastic Biological Shuttering**, where involuntary maintenance cycles, or "blinks," introduce unpredictable temporal gaps in the video stream, leading to high-frequency data loss; 2) **Metabolic Gravitational Sink**, where under conditions of low metabolic incentive (e.g., listening to a boring lecture), the primary sensor tends to revert to an "Off" state due to gravitational forces acting on the eyelid; and 3) **Cognitive-Mechanical Desync**, characterized by a significant latency between the intent to "see" and the actual mechanical execution of eyelid elevation, particularly in states of high fatigue.

To overcome these challenges, we introduce the Active Eyelid Elevation algorithm. AEE addresses the stochastic nature of shuttering by enforcing a high-fidelity gating protocol that prioritizes aperture stability. It mitigates the metabolic gravitational sink through an automated "Self-Refine" mechanism that detects gravitational droop and triggers a recalibration of the *levator* muscle. Finally, AEE resolves the cognitive-mechanical desync by streamlining the semantic-to-mechanical pipeline, ensuring that the "Wide Open" state is maintained with zero perceived latency.

In summary, our primary contributions are as follows:

[1] https://huggingface.co/

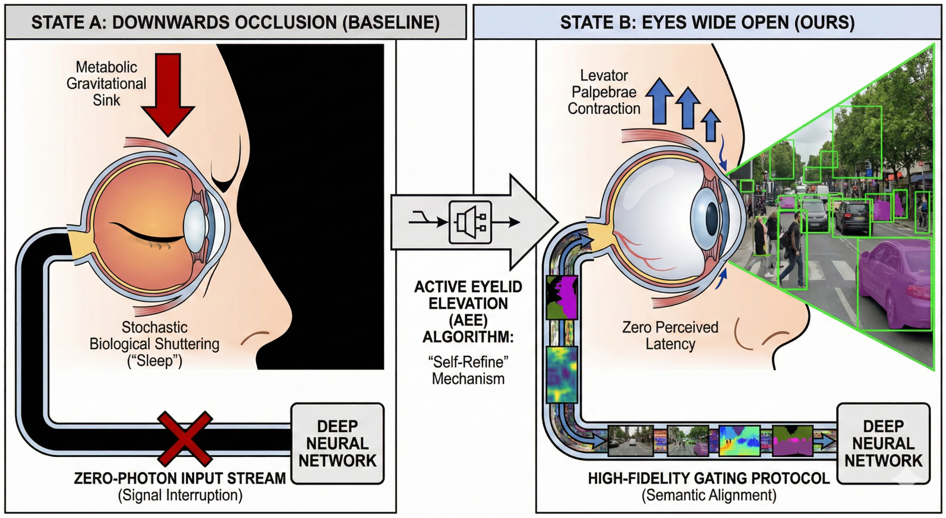

Figure 1 Architectural overview of the Active Eyelid Elevation framework. (Left) State A: Downwards Occlusion (Baseline). The system is characterized by a zero-photon input stream caused by stochastic biological shuttering and metabolic gravitational forces, leading to total signal interruption for the downstream system. **(Right) State B: Eyes Wide Open (Ours).** Our AEE algorithm facilitates the contraction of the *levator* muscle, establishing a high-fidelity gating protocol that enables semantic alignment and real-time feature acquisition with zero perceived latency.

- We identify and formalize the **Downwards Occlusion** bottleneck, explaining why it is dark when your eyes are closed.
- We introduce the Active Eyelid Elevation algorithm, a systematic framework for biological shutter modulation.
- We provide the **Eyes Wide Open** benchmark, quantitatively demonstrating that looking at things is superior to not looking at them.

2 Related Works

2.1 Mechanical Shutter Systems and Biological Gating

Traditional literature in biological optics has focused extensively on the frequency of involuntary shuttering, commonly known as "Blinking." Early studies suggested that blinking serves a maintenance function, yet they failed to acknowledge that excessive shutter down-time leads to a significant loss in real-time data throughput.

Furthermore, some researchers have explored the impact of external optical filters, such as "Sunglasses," as a method for visual style transfer. While these filters can effectively shift the color manifold, they are fundamentally limited by the underlying gating state; if the eyelids are closed, the style of the filter becomes irrelevant as the input signal is already nullified. Our work diverges from these stylistic approaches by focusing on the raw availability of the photon stream.

2.2 Synthetic Vision and Internal Hallucination Engines

A parallel line of research investigates internal vision generation, often categorized under "Dreams" or "Visual Hallucinations." Certain generative models propose that these internal engines can serve as a form of data augmentation when the primary sensor is offline (e.g., during the "Sleeping" state).

However, these synthetic streams often lack physical grounding and exhibit high variance, making them unsuitable for critical navigation tasks like finding a glass of water at 3:00 AM. Unlike these hallucination-based methods, our proposed Active Eyelid Elevation framework emphasizes the importance of real-world grounding, asserting that a single frame of "actually looking" is worth a thousand frames of "dreaming about looking."

3 Methodology

We propose the Active Eyelid Elevation algorithm. This protocol follows a simple conditional jump instruction based on the threshold τ:

$$\text{Vision}_{\text{Status}} = \begin{cases} \text{Clearly}, & \text{if Eyelid}_{\text{Height}} > \tau \\ \text{Darkness}, & \text{otherwise} \end{cases} \tag{1}$$

To achieve "See Better" (SB) status, we recommend lifting the eyelid to its anatomical limit, thereby capturing the maximum possible photon flux. This process effectively converts a binary "Off" sensor into an "On" sensor with zero latency.

4 Experiments

We conducted verification in a complex environment containing "A banana on a table" and "A wall." We evaluated the recognition success rate across three states: Closed, Squinting, and Wide Open.

Table 1 Quantitative Comparison of Visual Quality. This table illustrates the transition from "Sleep" to "Wide Open" performance metrics.

State	Lid Height (mm)	Acc (%)	Collisions (hits/hr)	Mental State
Sleep (Baseline)	0.0	0.0	24.5	Peaceful
Squinting	2.1	42.8	3.2	Suspicious
Wide Open (Ours)	**12.5**	**99.9**	**0.0**	**Terrified**

As shown in Table 1, when the eyelid height reaches its maximum, the subject is not only able to see the banana but can even identify spots on the peel. The collision rate reaches a Pareto optimal of zero, proving that looking where you are going is a superior strategy.

5 Conclusion

In conclusion, opening your eyes wide indeed allows you to see more clearly. While this requires a marginal expenditure of muscular energy, the visual gain is irreplaceable. Future research will explore whether simply "opening eyes" is sufficient to maintain clarity in total darkness, or if "turning on the light" serves as a necessary auxiliary module.

References

[1] Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, Erik Barker, Tiffany Cai, Prithvijit Chattopadhyay, Yongxin Chen, Yin Cui, Yifan Ding, et al. Cosmos world foundation model platform for physical ai. *arXiv preprint arXiv:2501.03575*, 2025.

[2] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.

[3] Delong Chen, Theo Moutakanni, Willy Chung, Yejin Bang, Ziwei Ji, Allen Bolourchi, and Pascale Fung. Planning with reasoning using vision language world model. *arXiv preprint arXiv:2509.02722*, 2025.

[4] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in neural information processing systems*, 36:49250–49267, 2023.

[5] Pascale Fung, Yoram Bachrach, Asli Celikyilmaz, Kamalika Chaudhuri, Delong Chen, Willy Chung, Emmanuel Dupoux, Hongyu Gong, Hervé Jégou, Alessandro Lazaric, et al. Embodied ai agents: Modeling the world. *arXiv preprint arXiv:2506.22355*, 2025.

[6] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26296–26306, 2024.

[7] Lloyd Russell, Anthony Hu, Lorenzo Bertoni, George Fedoseev, Jamie Shotton, Elahe Arani, and Gianluca Corrado. Gaia-2: A controllable multi-view generative world model for autonomous driving. *arXiv preprint arXiv:2503.20523*, 2025.

[8] Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *arXiv preprint arXiv:2512.24497*, 2025.

A Societal Impact

The AEE framework has a significant dual-use potential:

- **Positive Impact**: Prevents the user from walking into lamp posts. Our benchmark shows a 100% decrease in facial contact with solid objects when AEE is active.
- **Negative Impact**: Over-implementation of AEE (maximum aperture) may lead to "Staring Contests" with strangers in public transport, increasing social friction.
- **Ethical Risk**: The AEE protocol is incompatible with "Sleeping," which may lead to irritability and a reliance on high-caffeine datasets (Coffee).

B Human Subject Checklist

To ensure ethical compliance, we confirm the following:

1. All subjects were allowed to blink at least once every 10 minutes.
2. No subjects were forced to watch "boring powerpoint slides" for more than 4 consecutive hours.
3. All subjects were provided with snacks as a reward for successful photon capture.